

Step 1: Write the Given Network

- **Network Address:** 192.168.10.0
- **CIDR:** /24
- **Subnet Mask:** 255.255.255.0

A /24 network has **256 IP addresses** (from **192.168.10.0** to **192.168.10.255**).

Step 2: Find How Many Bits to Borrow

We need **4 equal subnets**.

Use the formula:

$$2^n = \text{Number of Subnets}$$

Where **n** = **borrowed bits**

- $2^2 = 4$

So, **borrow 2 bits**.

Step 3: Find the New Subnet Mask

A /26 prefix gives the subnet mask:

255.255.255.192

Step 4: Calculate the Block Size

Use this formula:

$$\text{Block Size} = 256 - \text{Last Octet of the Subnet Mask}$$

$$= 256 - 192$$

$$= \mathbf{64}$$

Subnet	Network Address	First Host	Last Host	Broadcast Address
Subnet 1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255